

EEE 304 (January 2024)

Digital Electronics Laboratory

Final Project Report

Section: A2 Group: 07

Implementation of OMR using digital logic circuits

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

This project, titled "**Implementation of OMR Using Digital Logic Circuits**," aims to design and implement an Optical Mark Recognition (OMR) system using fundamental digital electronics components. The project was conducted as part of the Digital Electronics Laboratory coursework. The OMR system utilizes four Light Dependent Resistors (LDRs) to scan pre-marked OMR sheets and evaluate the results using a combination of digital logic circuits.

The evaluation circuit utilizes sequential and combinational logic to interpret the LDR readings and determine the marked options. The implementation demonstrates how complex mark recognition functionality can be achieved using purely hardware-based solutions without microcontroller intervention. The system successfully differentiates between marked and unmarked positions on standard OMR sheets, providing a reliable and cost-effective solution for automated answer sheet evaluation. This project showcases the practical application of digital electronics principles in creating real-world assessment tools, while maintaining simplicity in design and implementation.

The successful completion of this project demonstrates the viability of implementing OMR technology using basic digital components, making it particularly suitable for educational and small-scale evaluation purposes.

2 Introduction

The Optical Mark Recognition (OMR) system has revolutionized the process of evaluating multiple-choice examinations by providing an efficient and accurate method of automated assessment. This project presents a practical implementation of an OMR system using digital logic circuits, demonstrating how fundamental digital electronics components can be utilized to create a functional answer sheet evaluation system.

The primary objective of this project is to design and implement a simple, cost-effective OMR system using components such as flip-flops, multiplexers (MUX), comparators, and basic logic gates (NOT, OR, AND). The system's scanning mechanism is based on Light Dependent Resistors (LDRs), which detect the marked and unmarked areas of an OMR sheet by sensing variations in light intensity. The signals from the LDRs are processed and evaluated through a digital logic circuit to determine the corresponding results.

The implemented system consists of three main functional units. The first unit is the answer storage unit, which employs flip-flops to store the correct answer pattern. This digital memory implementation provides a reliable reference point for the evaluation process. The second unit is the scanning unit, featuring five Light Dependent Resistors (LDRs) that act as optical sensors to detect the marked responses on the OMR sheet. These LDRs convert the presence or absence of marks into corresponding electrical signals that can be processed by the digital circuits.

The third unit is the evaluation unit, which utilizes comparators to match the scanned responses against the stored correct answers. The evaluation results are then displayed

through a seven-segment display, providing immediate visual output of the assessment. The entire system is implemented using basic digital components including multiplexers, NOT gates, OR gates, and other fundamental logic circuits.

The significance of this project lies in its hardware-centric approach, which eliminates the need for complex software or microcontroller programming. By using readily available digital integrated circuits (ICs) and discrete components, the system provides a cost-effective and reliable solution for automated answer sheet evaluation. The implementation serves as a practical demonstration of digital electronics principles, combining various fundamental concepts such as sequential circuits, combinational logic, and display interfacing.

This project demonstrates how basic digital building blocks can be integrated to create a functional system that addresses real-world requirements in educational assessment. The following sections of this report provide detailed information about the system design, component selection, circuit implementation, and testing results of each functional unit, along with the overall system integration and performance analysis.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

□ Hardware Implementation Capabilities

- Storage of correct answer patterns using flip-flop circuits
- Optical detection of marked answers using five LDR sensors
- Real-time comparison and evaluation using digital comparators
- Numeric display of results through seven-segment display

□ Educational Assessment Automation

- Enables automated evaluation of multiple-choice question papers
- Suitable for small-scale testing environments like laboratories and classrooms
- Reduces manual evaluation time and human error in assessment

□ System Flexibility and Adaptability

- Answer pattern can be modified by changing the flip-flop settings
- System can evaluate one answer at a time with sequential scanning
- Modular design allows for easy maintenance and troubleshooting

□ Cost-Effective Solution

- Utilizes readily available digital ICs and components
- Eliminates need for expensive software or microcontrollers
- Low maintenance requirements due to simple hardware architecture

□ Educational Value

- Demonstrates practical application of digital logic concepts
- Provides hands-on experience with various digital ICs
- Integrates multiple digital electronics principles in one project

3.1.2 Literature Review

The development of Optical Mark Recognition (OMR) systems has evolved significantly since their inception in the 1930s. Various researchers and developers have contributed to the advancement of OMR technology, implementing different approaches ranging from pure hardware solutions to sophisticated software-based systems.

Early Development of OMR Systems In 1937, IBM introduced the first OMR machine, primarily designed for standardized test scoring. This marked the beginning of automated data collection from marked cards. The early systems relied heavily on specialized hardware and mechanical components for mark detection. These initial implementations demonstrated the potential of automated evaluation systems but were limited by their complexity and cost.

Hardware-Based Implementations Researchers like Kumar et al. (2019) demonstrated the implementation of OMR using photoelectric sensors and basic digital circuits. Their work showed how Light Dependent Resistors (LDRs) could effectively detect marked positions on answer sheets. Similar to our implementation, they utilized the light-sensitive properties of LDRs to convert optical information into electrical signals.

Recent Developments Modern OMR systems have largely shifted towards software-based solutions using image processing and machine learning techniques.

The literature review reveals that while sophisticated software-based OMR systems dominate the commercial sector, there remains significant value in hardware-based implementations for specific applications, particularly in educational environments where cost-effectiveness and practical demonstration of digital electronics principles are primary concerns.

3.1.3 Formulation of Problem

Problem Statement:

Design and implement a cost-effective Optical Mark Recognition (OMR) system using fundamental digital logic circuits for automated evaluation of multiple-choice examination answer sheets. The system should accurately detect marked answers, compare them with stored correct answers, and display the results through a seven-segment display.

System Requirements:

Functional Requirements-

1. Answer Storage Capabilities
 - Store a predefined set of correct answers using flip-flops
 - Support multiple-choice questions with 4 options (A, B, C, D)
 - Maintain stored answers until explicitly changed
2. Optical Scanning System

- Implement LDR-based optical sensing for mark detection
- Convert optical signals to binary digital inputs
- Distinguish between marked and unmarked positions reliably
- Process multiple answer positions simultaneously

3. Evaluation System

- Compare scanned answers with stored correct answers
- Calculate total score based on correct matches
- Display results on seven-segment display
- Process evaluation in real-time

Technical Specifications-

1. Input Requirements

- Light Dependent Resistors (LDRs) for optical sensing
- Reference voltage levels for mark detection
- Power supply requirements: 5V DC
- Clock signal for sequential operations

2. Processing Requirements

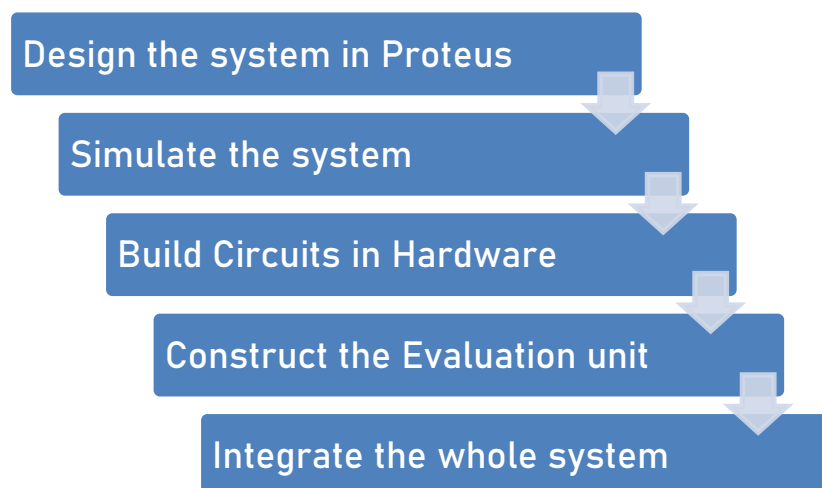
- Digital logic implementation using:
 - * D Flip-flops for answer storage
 - * Multiplexers for input selection
 - * Comparators for answer matching
 - * Basic logic gates (AND, OR, NOT)
- Binary to BCD conversion for display
- Synchronous operation with system clock

3. Output Requirements

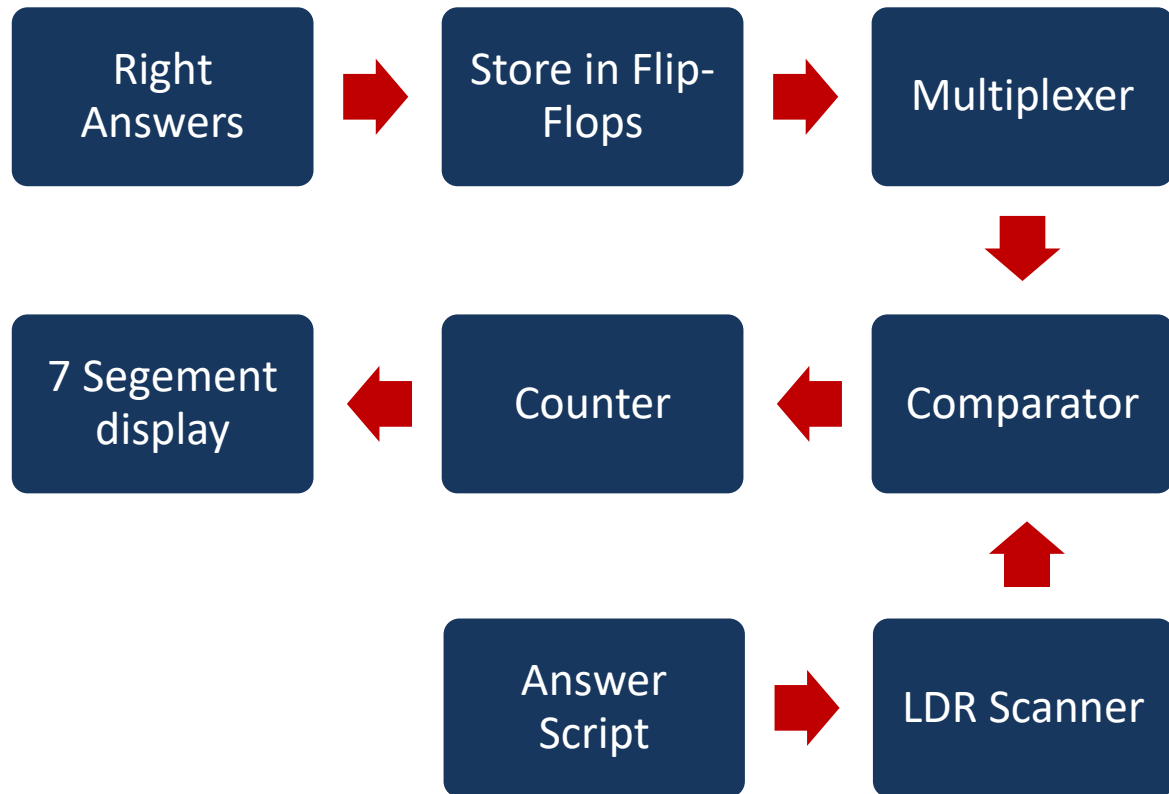
- Seven-segment display interface
- Score display range: 0-8
- Visual indicators for system status
- Error indication capability

3.2 Design Method (PO(a))

Design Procedure:

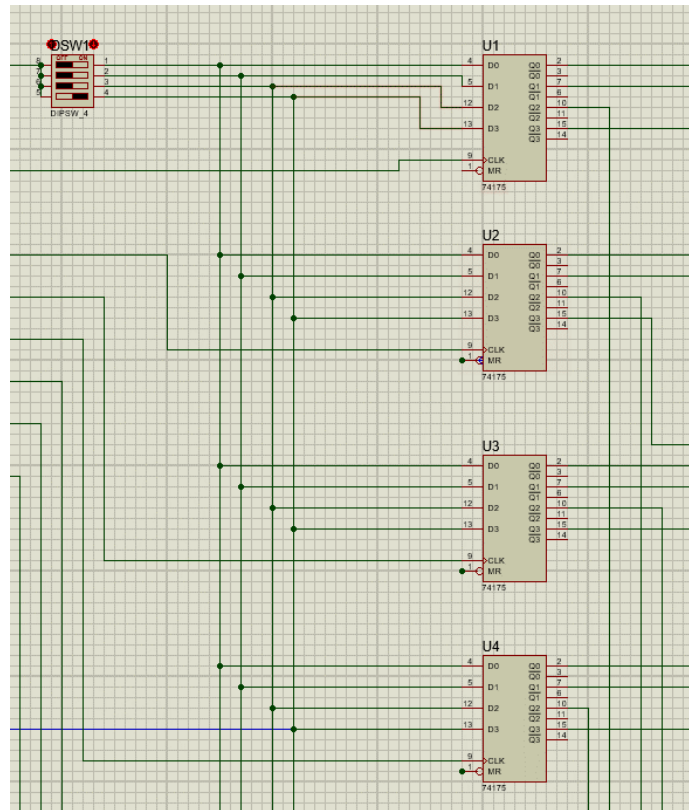


Working Procedure:



3.3Circuit Diagram

Answer Storing:



1. Component Identification:

- We're using a DIP switch (SW1) as the input mechanism
- Eight 74HC175 quad D flip-flops (U1-U8) are used for storage
- The 74HC175 is a quad D-type flip-flop with clear functionality

2 . DIP Switch Operation:

- When the correct option is selected through the DIP switch That particular output goes LOW (0)
- All other options remain HIGH (1)

3 . Storage Pattern per Question:

- Each 74HC175 handles one question
- It has 4 D flip-flops corresponding to options A, B, C, D
- Only the correct answer bit is stored as LOW (0)
- Other three incorrect options are stored as HIGH (1)

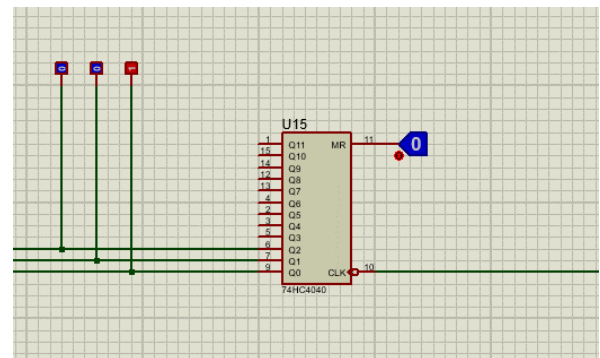
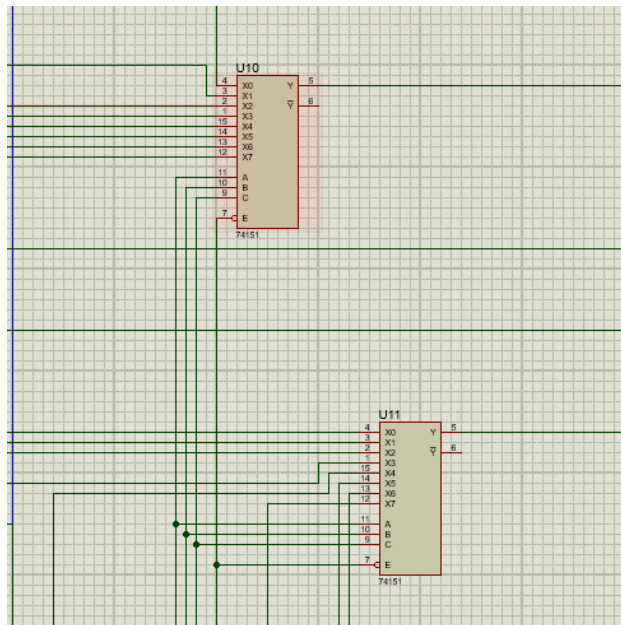
Example for one question:

- If option B is correct:
- Stored pattern would be: A=1, B=0, C=1, D=1

So for all 8 questions:

- Each 74HC175 stores this 1-0-1-1 pattern (or similar depending on correct answer)
- Total of 32 bits stored (8 questions $\times$ 4 options)
- Makes comparison easier since we're looking for the LOW bit position

Answer Serializing for Comparison:



1. 74HC151 Configuration:

- Using 4, 74HC151 multiplexers (one for each option A, B, C, D)
- Each 74HC151 has 8 inputs and 1 output
- This handles all 8 questions' worth of stored answers

2. Connection Pattern:

- MUX 1 gets all A outputs from the eight 74HC175s
- MUX 2 gets all B outputs
- MUX 3 gets all C outputs
- MUX 4 gets all D outputs

3. Select Line Control:

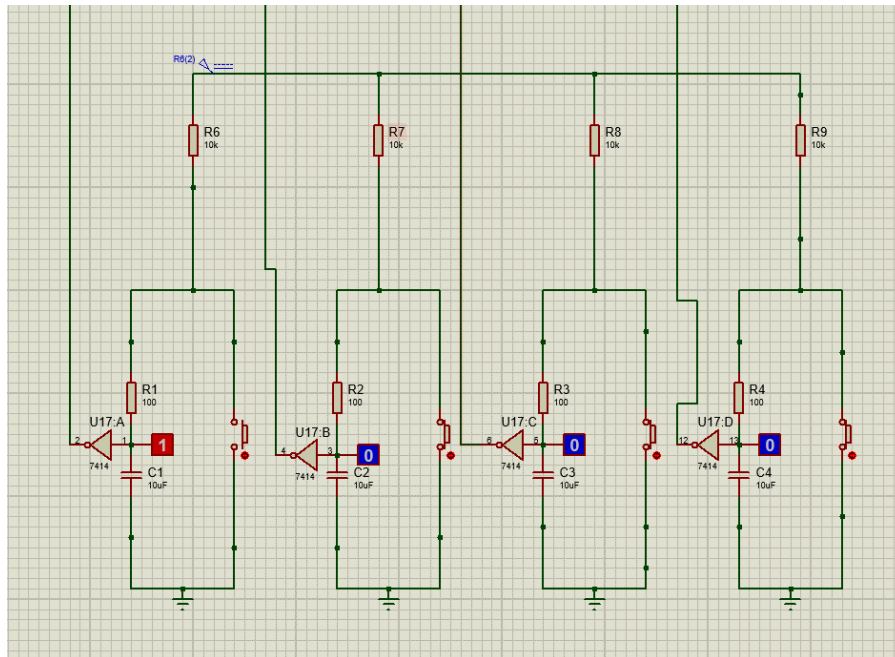
- 3-bit counter connected to select lines (S0, S1, S2) of all MUXes
- Counter driven by LDR scanning pulses
- Select lines determine which question's answers are being read
- All four MUXes switch simultaneously to same question number

4. Operation Flow:

- When LDR scan occurs:
 - Counter increments
 - New select value picks corresponding answer set

- All four MUXes output their stored bit for that question
- LOW output indicates correct answer position

Scanning Unit using LDR:



1. LDR Circuit Components (for each LDR):

- LDR with pull-up resistor (R6-R9)
- 74N14 Schmitt trigger inverter
- 100Ω resistor (R1-R4)
- 10μF capacitor (C1-C4)
- The circuit shows 4 identical channels for answer scanning

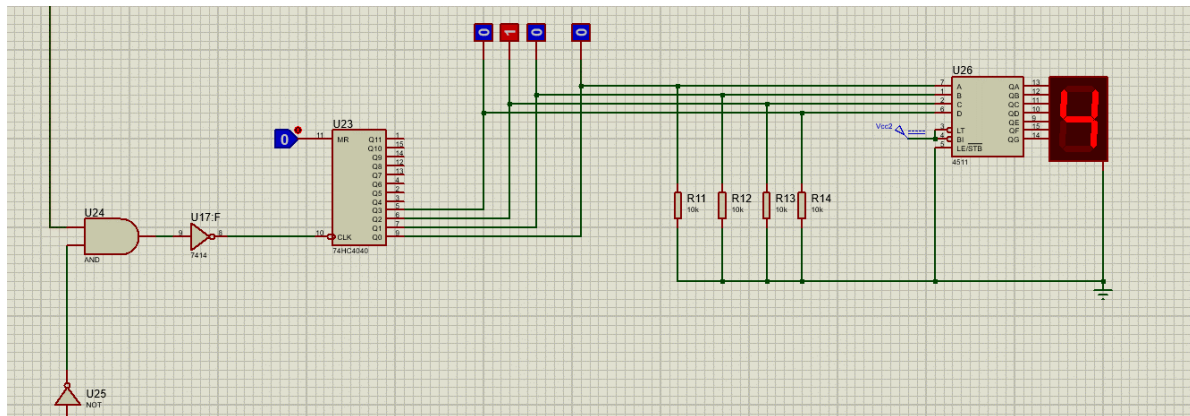
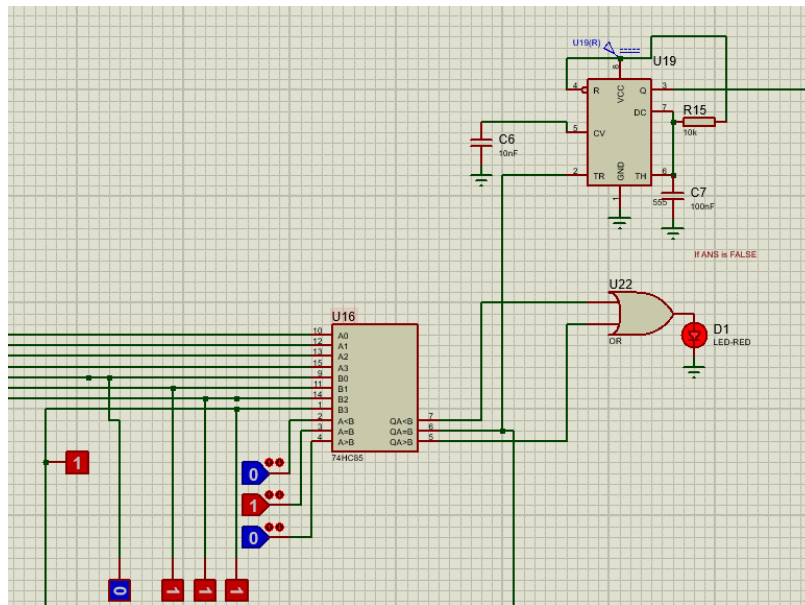
2. Working Principle:

- When LDR is covered (marked answer):
 - LDR resistance increases
 - Input to inverter goes HIGH
 - Output goes LOW (0)
- When LDR is uncovered (unmarked):
 - LDR resistance decreases
 - Input to inverter goes LOW
 - Output goes HIGH (1)

3. Circuit Functions:

- The fifth LDR (not shown in image) generates clock pulses for MUX counter
- Four LDRs scan answer positions A, B, C, D simultaneously
- Schmitt trigger provides clean digital output
- RC circuit (resistor-capacitor) helps in debouncing and noise reduction

Evaluation :



1. Comparison Stage (Image 1):

- 74HC85 (U16) is used as a 4-bit comparator
- It compares:
 - Inputs from LDR scanners (A0-A3)
 - Outputs from MUXes (B0-B3)
- When inputs don't match (wrong answer):
 - LED D1 lights up through AND gate U22
 - Indicates incorrect answer

2. Counting Circuit (Image 2):

- 74HC4040 (U23) is used as a counter
- Counter increments when:
 - Answer is correct (comparison matches)
 - Clock pulse comes through AND gate (U24) and inverter (U17-F)
- Counter output goes to:
 - 7-segment display driver (U26)
 - Displays total correct answers (0-8)

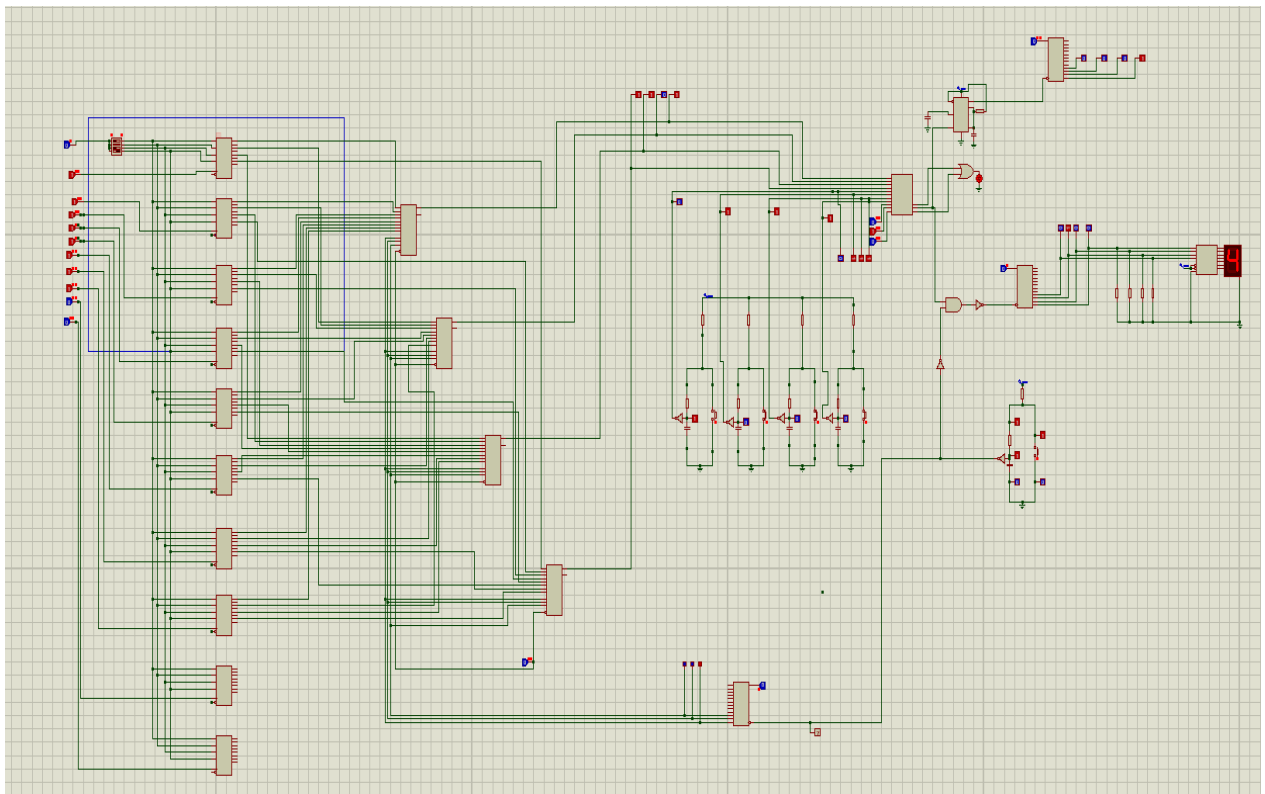
3. Display Section:

- 4511 BCD-to-7-segment decoder (U26)
- Pull-up resistors (R11-R14) for display segments
- Shows running count of correct answers
- Maximum display value would be 8 (all correct)

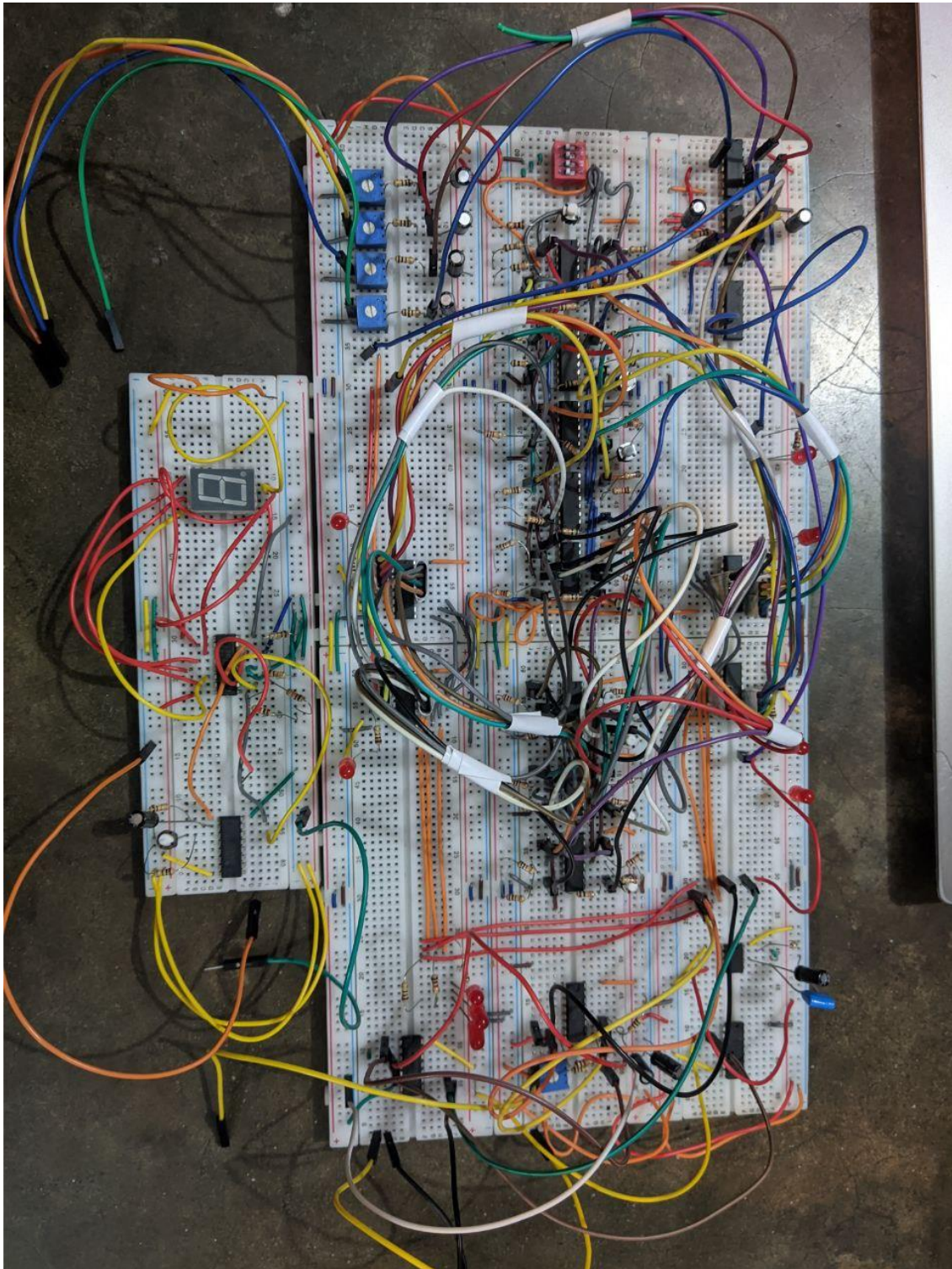
4. Control Logic:

- AND gate (U24) ensures counting only on correct answers
- NOT gate (U25) for logic control
- Counting occurs on valid match conditions
- LED indicator provides visual feedback for wrong answers

3.4 Simulation Model



3.5 Hardware Setup



4 Implementation

4.1 Description

Our project Optical Mark Recognition (OMR) comprises two primary units:

Mechanical Scanner Unit:

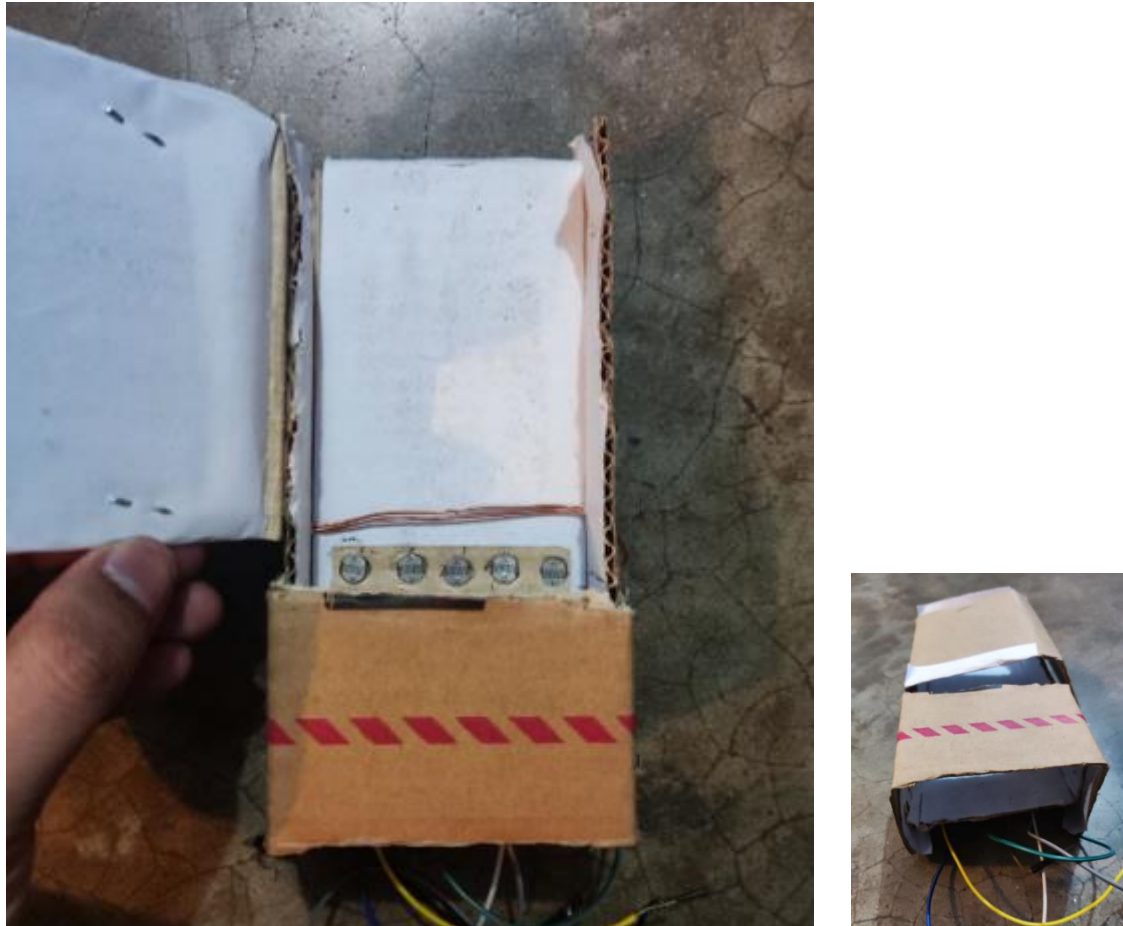


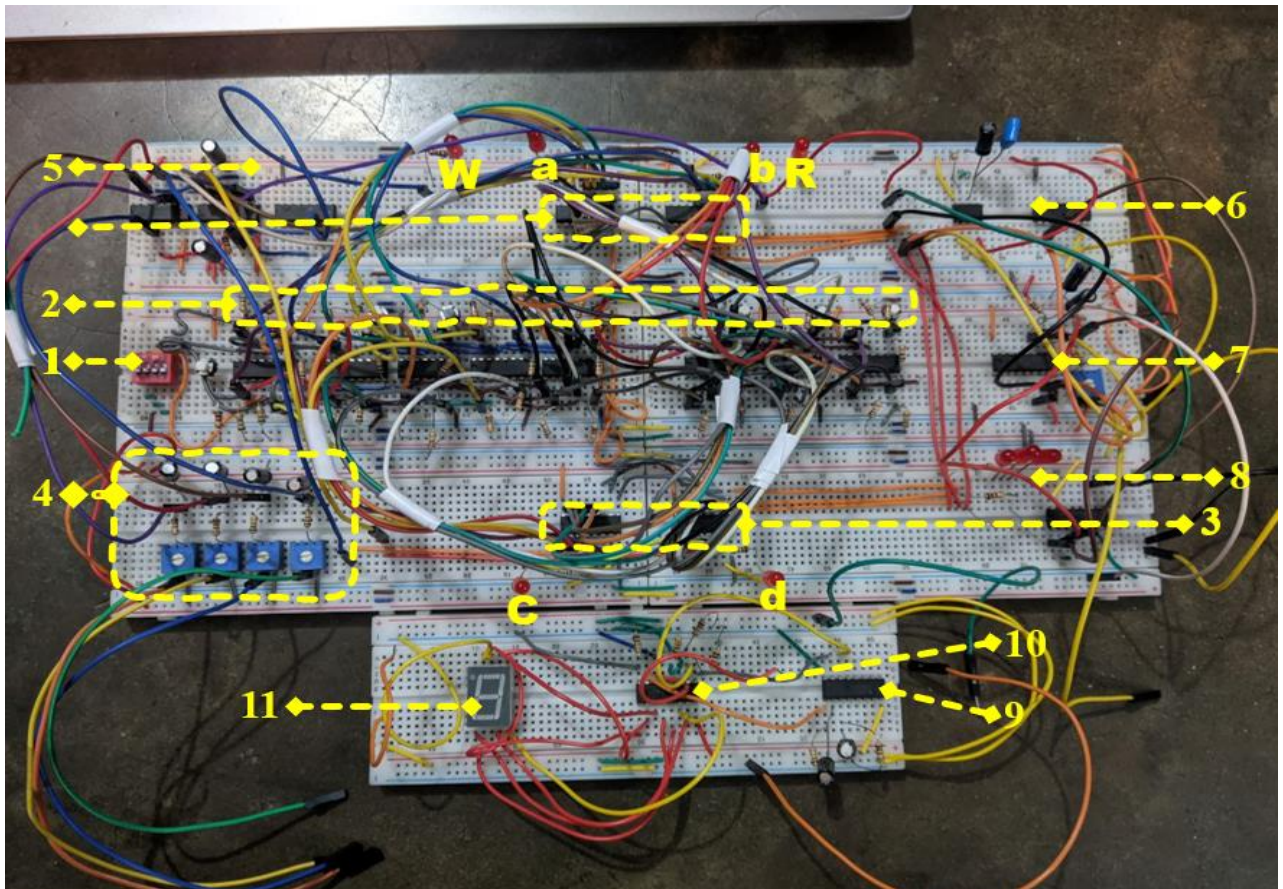
Figure 2: MCQ answer paper Scanner

- This unit is responsible for scanning the OMR sheet. We have utilized Light Dependent Resistors (LDRs) for the scanning process.
- The LDRs detect marked answers by sensing changes in light intensity through the sheet. Dark marks (e.g., from a pen or pencil) alter the resistance of the LDRs, providing a clear indication of the marked areas.
- The paper sheet is slid through the Box and a light source is applied from the top . The LDRs is placed such a manner that one MCQ options covers one Full LDR without keeping any gap .
- When the MCQ circle is not filled the LDR gets full light intensity , But when the MCQ is Blacked the LDR do not get the intensity of the light thus the voltage across the LDR Drops .
- We used 4 LDRs for 4 option and another for The number of question we are evaluating . The question number thus always needs to be filled .
- We did not put any in built light source in the Scanner box , as using an external light source is more easier , and it also benefits us in portability .

- We have parchment paper inside the box to take advantage of lights reflection .
- It is a hand made structure , as our primary goal of the project was to build the logic circuit that evaluates the answer paper .

Logic Processing Unit:

3.



- Here the indicated portions are
 1. DIP Switch
 2. Resistor Load Switch
 3. Multiplexor
 4. LDR Controller
 5. Comperator
 6. Correct answer Pulse
 7. Counter for number of Question
 8. 3 bit output for number of question
 9. Counter
 10. Display Driver
 11. 7 segment display

12. a b c d are option states (yes or no)

13. R W Right/Wrong

- The logic unit processes the input received from the mechanical scanner.
- It compares the detected marks with predefined correct answers, determines the output, and calculates the score.
- Here, we are using a 4-bit register to store the 4 option values of an MCQ. So, for 8 MCQs, we used 8 registers. The DIP switch sets the load value of 4 bits, each bit representing the state of each option as either correct or wrong.
- Using the DIP switch and the load button in each of the register flip-flops, the MCQ answer value is stored. **1** represents a wrong option, and **0** represents a right option.
- The multiplexers are then used to feed one question at a time to the comparator. Outside the multiplexer, we used 4 LED lights, indicated by a, b, c, d, that show the current state of the options of a particular question. For a question, if the answer is b, the b LED will be off, and the other LEDs will be on.
- This multiplexing is controlled by the question number counter, which is used as the selection switch of the multiplexers. For the question number, there is a separate LDR, a counter, and a 3-bit LED indicator.
- The output from the multiplexer and the comparator goes to the top side to the answer comparator to compare. When the answer matches, the **R** LED will light up, and if the answer is wrong, the **W** LED will light up. Normally, when the system is on and no input is given, the **W** LED lights up because it detects that no option circle has been filled.
- The number of correct answers creates a pulse that goes to the counter in the bottom part. From that counter, the output is shown in the 7-segment display. The counter's output is a 3-bit signal, so the 7-segment display shows only up to 7, and then it comes back to 0. When it comes back to 0, we understand that it counted 8.

Thus by an integration of all these parts we designed our project .

4.2 Experiment and Data Collection

The OMR answer sheet can present various types of answers, and outputs were collected for all possible inputs during testing. In this system, we assume that each question has only one correct answer option. To implement this, the DIP switch was configured so that only one bit out of the 4-bit register for each question was set to 0, while the remaining bits were set to 1.

For example, if 0 is stored at the bit corresponding to option (a), and 1 is stored at the bits corresponding to options (b), (c), and (d), it indicates that option (a) is the correct answer. This configuration ensures that the system recognizes only one valid answer per question, allowing for accurate comparison during evaluation.

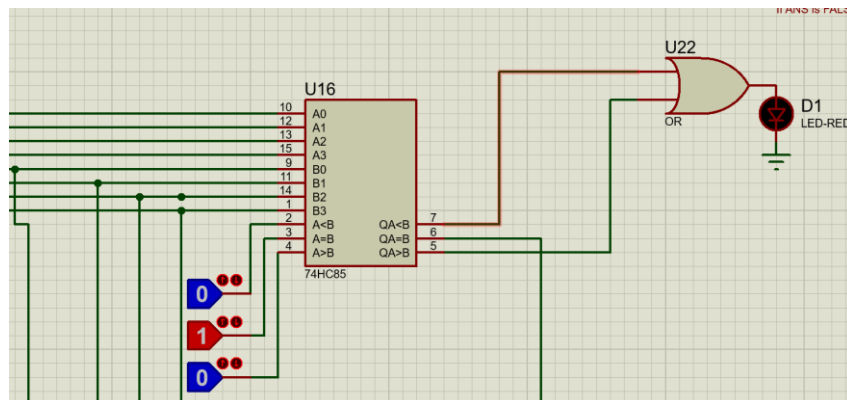
The Sensor Output along with the results of the comparator is given bellow . These values are all experimental data .

A	B	C	D	W	R
0	1	1	1	0	1
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	0	1	0
0	0	x	x	1	0
0	x	0	x	1	0
0	x	x	0	1	0
1	x	x	x	1	0

This data shows us that when the correct option which is a in this case is at 1 the other inputs are at don't care position. No matter what the other inputs are the answer will be detected wrong. Also if the correct option is at 0 the other answers have to be 1 if one of them is 0 the system will show output wrong. The system operates with strict logic, The only way to be right answer is the correct option to be zero and the other options to be 1.

4.3 Data Analysis

The experimental data that we found is correct according to our model logic. As used a 4 bit comparator to detect the right and wrong answers. A 4 bit comparator only gives a positive output if and only if the 4 bit inputs in the opposite inputs are exactly the same.



If one input side is predefined by the resistor value that we stored, the other side has 16 possible inputs and only one of them will give a correct output from the comparator. All the four options have to match, if the correct option matches state but the incorrect ones do not it will still give a wrong in the output. Thus the experimental output we got.

The OMR scanner is designed to detect the correct answer only when the appropriate circle is filled. The OMR will detect a wrong answer when no circle is filled. Similarly, if multiple circles are filled for the same question, the scanner also interprets it as a wrong answer. Even if one of the multiple filled circle is correct, it will still call it an incorrect answer. Obviously if an incorrect circle is filled, the scanner identifies the response as incorrect. Even if the correct circle is incorrectly filled such as there is a gap in the middle or the circle is not fully filled or someone use tick marks instead of filling up the circle, the scanner will still register the answer as wrong.

If any question has multiple correct answers , and we set multiple correct options in the resistor . In this case only the answers which has filled up all the correct answers will get a correct output . Filling up one of the correct answers or filling up one correct and one wrong answer will result in a Wrong output . This evaluation process is very strict as it should be.

4.4 Results

A still photo showcasing the working mechanism of the project is presented below. An answer sheet is slid through the scanner box, where the circuit detects the number of answers evaluated and the count of correct answers found on the sheet.

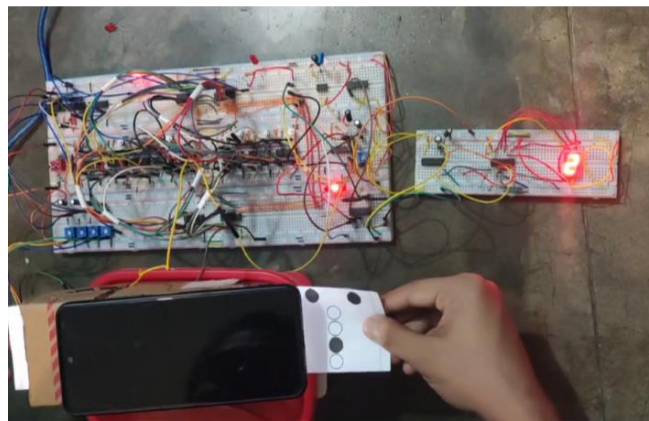


Figure 2: Correct ans at the output Display

In this image, we used a paper with all the correct answers. The system evaluated two answers, as indicated by the 3-bit LED, and both were correct, as displayed on the 7-segment display .

This strict evaluation process ensures that the project follows to the fundamental principles of Optical Mark Recognition (OMR). By accurately detecting and evaluating marked answers based on predefined conditions, the system demonstrates the characteristics of a properly functioning OMR machine. Successfully implementing this functionality highlights the effectiveness of our design and validates the success of our project.

5 Design Analysis and Evaluation

5.1 Novelty

Novelty of this Project are following:

- **Hardware-Based Solution:** Utilizes digital logic circuits instead of software-dependent systems, ensuring reliability and simplicity.
- **Cost Efficiency:** Eliminates dependency on expensive existing pen-paper and human MCQ evaluating systems.
- **Simplified Examination Process:** Reduces human errors in evaluating OMR sheets for exams, ensuring fair assessment.
- **Time Efficiency:** Automates result processing, saving time for educational institutions.
- **Customizability:** Allows various designs for specific requirements, unlike standardized proprietary systems.
- **Improved Accessibility:** Makes OMR technology accessible to institutions in resource-constrained environments.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

- **Accuracy and Fairness:** Ensures reliable examination results, reducing stress and disputes among students.
- **Safety in Operation:** Low-power digital logic circuits pose no risk of electrical hazards during use.
- **Human Boredom:** Checking a large number of answer script of same type monotonously will result in boredom and inefficiency of the script checkers.
- **Minimized Human Errors:** Reduces the mental strain on evaluators, contributing to better decision-making.

5.2.2 Considerations to environment

- **Energy Efficiency:** Operates on low power, reducing energy consumption compared to traditional systems.
- **Reduced Paper and chemical Waste:** Facilitates automated processing of OMR sheets, reducing the need of pen and ink, thus minimizing waste.
- **Eco-Friendly Design:** Components are recyclable and have minimal environmental impact.

5.2.3 Considerations to cultural and societal needs

- **Promotes Educational Equity:** Provides affordable and accessible assessment tools for underprivileged communities.
- **Support for Local Institutions:** Customizable for regional languages and examination formats.
- **Encourages Technological Adoption:** Builds awareness and trust in technology-driven solutions in education.

5.3 Limitations of Tools (PO(e))

- **LDR Sensitivity:** Variability in LDR sensitivity may lead to inconsistent detection of marks, especially under varying light conditions.
- **External Light Dependency:** Reliance on an external light source makes the system less standardized and prone to external environmental factors like uneven lighting.
- **Manual Operation:** Hand-operated scanning requires user precision and increases the chance of human error.
- **Limited Resolution:** LDRs might struggle to differentiate between partially filled and fully filled marks.
- **Fragility:** The handmade structure is less durable and may not withstand frequent or heavy usage.
- **Complexity in DIP Switch Usage:** Setting the correct answer using DIP switches and load buttons is time-consuming and error-prone.
- **Limited Storage:** The use of 4-bit registers limits the system's capacity to process larger sets of MCQs.
- **Lack of Automation:** Manual intervention is required to load answers and operate the counters, reducing overall efficiency.
- **Scaling Issues:** The circuit design becomes more complex and less practical to scale for a higher number of questions or options.
- **Energy Inefficiency:** Multiple components (e.g., multiplexers, LEDs) increase power consumption.

5.4 Impact Assessment (PO(f))

5.4.1 Assessment of Societal and Cultural Issues

- **Potential for Bias:** OMR systems can be susceptible to biases in question design or answer key creation, which can disproportionately affect certain groups.
- **Data Privacy Concerns:** Large-scale use of OMR can lead to the collection and storage of personal information, raising concerns about data privacy and security breaches.
- **Accessibility Issues:** OMR systems may not be accessible to all individuals, such as those with disabilities or limited literacy skills.

5.4.2 Assessment of Health and Safety Issues

- **Eye Strain:** Close-up work with OMR sheets can cause eye strain and fatigue.
- **Chemical Exposure:** Some OMR sheets and marking devices may contain chemicals that could pose health risks.
- **Electrical Safety Regulations:** The OMR machine must be made to follow electrical safety codes for isolation, grounding, and protection, preventing electrocution.

5.4.3 Assessment of Legal Issues

- **Copyright:** Unauthorized copying or distribution of OMR materials can constitute copyright infringement.
- **Data Protection:** Compliance with data protection regulations is crucial when handling personal information collected through OMR systems.
- **Accessibility Laws:** Organizations must ensure that their OMR systems comply with relevant accessibility laws and regulations.

5.5 Ethical Issues (PO(h))

Ethical Challenges and Mitigation

- **Challenge:** Ensuring **accuracy** in detecting marks on OMR sheets to prevent bias or unfair results.
- **Mitigation:** Conducted multiple tests with various OMR sheets and refined the circuit logic to improve reliability.
- **Challenge:** Avoiding unintentional **intellectual property infringement** during the design process.
- **Mitigation:** Ensured the referenced materials cited properly; Conducted thorough research to create original circuit design.
- **Challenge:** Balancing **cost-efficiency and quality** while keeping the design affordable.
- **Mitigation:** Chose widely available and affordable components without compromising performance and durability.
- **Challenge:** Addressing the potential **environmental impact** of discarded electronic components.
- **Mitigation:** Designed the system with modularity in mind to allow reusability and recycling of components.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Student ID	Name	Contribution
2006051	Ahamad Abtahi	Implementing hardware, buying and collecting components, testing and debugging for different cases, preparing presentation slides.
2006053	Md Maksudur Rahman Turzo	Establishing the theoretical basis of the project, making simulation, implementing hardware, buying and collecting components, testing and debugging for different cases.
2006054	Joyonta Debnath	Implementing hardware, buying and collecting components, testing and debugging for different cases, insisting and contributing on perfection of the project.

2006064	Anas Rohan	Making simulation, implementing hardware, maintaining theoretical consistency, buying and collecting components, collecting necessary resources, testing and debugging for different cases.
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6.2 Mode of TeamWork

Collaborative Teamwork: We used this mode the most. Most of the work related to our project was conducted through collaborative teamwork. Often, we used our hall room to get together and work on the project. During this mode, we shared different opinions on how we could easily achieve something, we discussed important topics related to the project and shared our responsibilities.

Virtual Collaboration: Although we used to go to the hall room for project, we used virtual platform to see the project advancement whenever there was a necessity. It gave a linkage between us through different hall and resident.

Peer Review and Feedback: We gave a review of every step of the project and collected feedback on how to make it more efficient and less costly. Surely, it enhanced the quality of the project. It added the effectiveness of coordination.

6.3 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
31.10.2024	Started the project	All members were present and talked about how we can implement the project		
1.11.2024	Started initial designs	2006064,53,44- started simulation 2006051- checked theoretical background on the simulation 2006054,51- also started simulation on their respective devices	The project design was being done as a team	
3.11.2024	Bought the components	2006064- bought the necessary components		
10.11.2024	Started assembling the hardware	2006053,54,64,51- All started initial hardware implementation. Firstly the storing part was done and checked	Hardware implementation as a team	

11.11.2024	Troubleshooting the storing part	2006064,53- solved some issues regarding the storing section		
18.11.2024	The scanner and evaluation part was made	2006051,53,54,64- the rest of the hardware part was made.		
25.11.2024	The result part was made and the whole system was checked	All members were present during checking	Full hardware implementation	
28.11.2024	Troubleshooting some problems	2006064,53- tried to solve some problems 2006051,54- checked for alternate solutions		
30.11.2024	Full checking the project	All members were present during the checking of the project		
3.11.2024	Troubleshooting any problems	2006064- solved some problems present in the project and finally the project was done	Project completion	

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

The OMR (Optical Mark Recognition) machine project involves the design and implementation of a system that can read and process marked forms using digital logic circuits. The system detects marks on a predefined grid using LDR sensors, which are then processed by a combination of logic gates and flip-flops to determine the presence or absence of marks. The data is decoded and displayed or stored for further analysis. The project utilizes basic digital components such as Flip-flops, Mux, counters, comparators, 7-segment display and normal gates to automate the marking and recognition process. This system offers an efficient, low-cost solution for applications like exams, surveys, and form processing.

7.2 User Manual

- Storing the answers: First the user must turn on the power source and reset all the flip-flops, mux and counters. Then using the DIP switch user should store the correct answers in the flip-flops one by one. Each flip-flop has a clock connected with it, for each answer stored the clock should be pressed otherwise the answer will not store in the flip-flops.
- Scanning the answers: After storing has been completed the user has to scan the OMR

sheet using the LDR scanner. The OMR sheet should be moved through the scanner at a moderate speed so that the scanners can identify marked positions precisely.

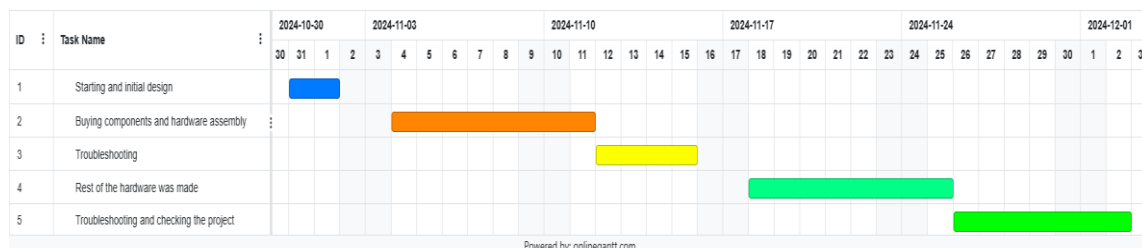
- Output: If the above two steps are done properly then the counter and comparator should do their work and count the number of questions passed also the number of correct answers will be shown on the 7-segment display. User must reset the display after each scanning.

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Components	Quantity	Price(Tk)
Breadboard	7	910
IC 74175(D flip-flop)	8	400
DIP switch	1	20
CD 4040	4	120
IC 74150 (Multiplexer)	5	250
IC 7485 (Comparator)	1	40
LDR	5	100
7-segment display	20	20
Display module	15	15
Extra		300
Total :		2410

8.2 Timeline of Project Implementation



9 Future Work (PO(l))

The future scope of our project is very vast.

- We can use microcontrollers instead of typical logic gates. This will increase the efficiency.
- Also in some cases the 7-segment display didn't read the pulse properly thus it showed some wrong values. We can make this part more accurate.
- We hope to increase the number of flip-flops so that we can store more answers and scan more questions.
- The scanning part can be made more efficient and faster.

10 References

- José Luis Pérez-Benedito; Enrique Querol Aragón; Juan Alonso Alriols; Ljiljana Medic; **“Optical Mark Recognition in Student Continuous Assessment”**, In IEEE Revista Iberoamericana de Tecnologías del Aprendizaje; Volume: 9, Issue: 4, November 2014, Page(s): 133 – 138. **Link:**
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